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Software Requirement Specification [SRS] Document On

Project Name: Pharma Track—Pharmacy Stock and Sales Management System [PSSMS]

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1. INTRODUCTION

1.1. Background Problem:

At present, most of the pharmacies are being organized by maintaining old traditional rules. Most of the pharmacy stake holders are dependent on pen-paper for recording, paper's file for tracking. Phone calls are the main method for contacting suppliers and customers. These types of manual activities slow down pharmacies' daily activities. When any important information needed, it's very hard to find, even sometimes, might not be found. Errors in billing, stock counting, record keeping are much familiar for these types of pharmacies. On the other hand, stakeholders expect a faster, reliable, easy-to-understand system. Traditional management cannot meet the needs of the modern digital pharmacy era.

In the present much-advanced digital era, the pharmaceutical industry should not lag behind. Pharmacy management has become quite an essential element for maintaining smooth operations and ensuring the delivery of quality health care services. Many small- and medium-sized pharmacies still rely heavily on manual systems, which quite often lead to mismanagement in their operations. Frequent errors such as stockouts, overstocking, and incorrect sales tracking reduce overall profitability. More importantly, these issues may compromise patient safety when critical or lifesaving medications are unavailable at the right time.

A pharmacy nowadays has to coordinate with several suppliers, accurately track batches, and monitor the expiration dates of drugs while being compliant with regulatory standards. These are all time-consuming activities if managed manually and prone to human error. Consequently, there is an increasing demand for a contemporary, integrated, automated system that would take care of such complicated processes.

Pharma Track - Pharmacy Stock and Sales Management System [PSSMS], a digital pharmacy management system, is designed to meet these challenges. It is an overall digital solution that equips pharmacies with effective tools for inventory management, real-time sales tracking, and expiration deadline tracking.

It will help the pharmacy to improve operational efficiency, remain compliant, and ultimately offer better healthcare services to customers and patients alike.

Key benefits of Pharma Track-PSSMS:

- Improved accuracy in inventory management
- Real-time sales tracking and data-driven insights
- Tracking upcoming expiries automatically with timely notifications
- Reduction in human errors and workflow bottlenecks
- Improved Compliance with Pharmaceutical Standards

Ultimately, Pharma Track-PSSMS will enhance the operational effectiveness of pharmacies, reduce financial losses, and ensure a constant supply of key medicines. It is high time to forget old traditional systems and rely on digital pharmacy management like PSSMS. A system like this can enhance pharmacy management in modern era.

1.2. Target Group of Users:

The Pharma Track-PSSMS summarizes information from all activities transacted within the system into useful insights. While the admin drives this reporting aspect of the use case, the information is supported by contributions provided by the five key identified user groups. This ensures that every operational action, every action related to inventory, sales, billing, and delivery, is indeed reflected in those final reports.
The reporting module becomes powerful because each of the five targeted users contributes essential data. The five targeted users of Pharma Track - Pharmacy Stock and Sales Management System [PSSMS] are:

- **Admin:** Admin is the head user here in PSSMS. He can control and check accuracy of each user. Admin will generate and review comprehensive reports, which include sales summaries, inventory status, listings for expiry, performance of suppliers, financial logs, and reports on user activities. Admin can update their details, track and sack other users.

- **Pharmacist:** Pharmacists have the second highest priority among the users of PSSMS. He will be appointed by admin. Pharmacists will provide the critical data through medicine updates, expiry monitoring, stock checks, and sales entries, which form the basis of inventory and expiry-related reports.

- **Cashier:** The main responsibility of cashier is to make up-to-date financial matters. Cashiers will provide critical data through medicine updates, expiry monitoring, stock checks, and sales entries, which form the basis of inventory and expiry-related reports. Even if some products are being imported or exported, he will be responsible for updating the financial calculation.

- **Supplier:** Suppliers know better about invoices and pricing. It will also ensure transparency. Suppliers will add delivery update information, batch numbers, invoices, and pricing information to contribute to the procurement and supply chain reports.

- **Customer:** Customers will be field-level users of PSSMS. They will be able to order inventory, check prices, expiration date, availability in stock, batch numbers and other info about pharmaceutical elements. They don't need to come to pharmacy to know about these, they will be able to know from home if they use PSSMS.

These are the five targeted users of PSSMS. It ensures completeness and reliability in the reporting system by integrating the various contributions from the five users. This smooth flow of information within the system improves accuracy, enhances transparency, and supports better decision-making across the pharmacy. At the end, the Generate Reports feature strengthens operational control and accountability and helps to maintain a well-organized and compliant pharmacy management environment.

2. SCOPES & FEATURES

2.1 Scope 1: Profile Management

Features:

- Store officers and customers can create new profiles with valid information.
- All users can log in with a valid username and password.
- After successful login, all users can view their dashboard.
- All users can view their own profile details.
- All users can update their information with valid information.
- All users can change their password.
- All users can recover their password after verification.

2.2 Scope 2: User Management (Admin Only)

Features:

- Admin can add new users (pharmacist, cashier, supplier).
- Admin can update user information.
- Admin can delete users from the system.
- Admin can assign roles and permissions to users.

2.3 Scope 3: Inventory Management

Features:

- Pharmacists can add new medicines with batch numbers, expiry dates, and pricing.
- Pharmacists can update existing medicine details.
- Admin can approve stock updates from pharmacists.
- Users can check stock availability.
- System can generate alerts for low stock and upcoming expiry dates.

2.4 Scope 4: Sales and Billing Management

Features:

- Cashiers can create bills and add items.
- Cashier can generate invoices for customers.
- Cashier can record payments.
- Cashier can submit daily sales reports.
- System can validate sales amounts automatically.

2.5 Scope 5: Reporting and Dashboard

Features:

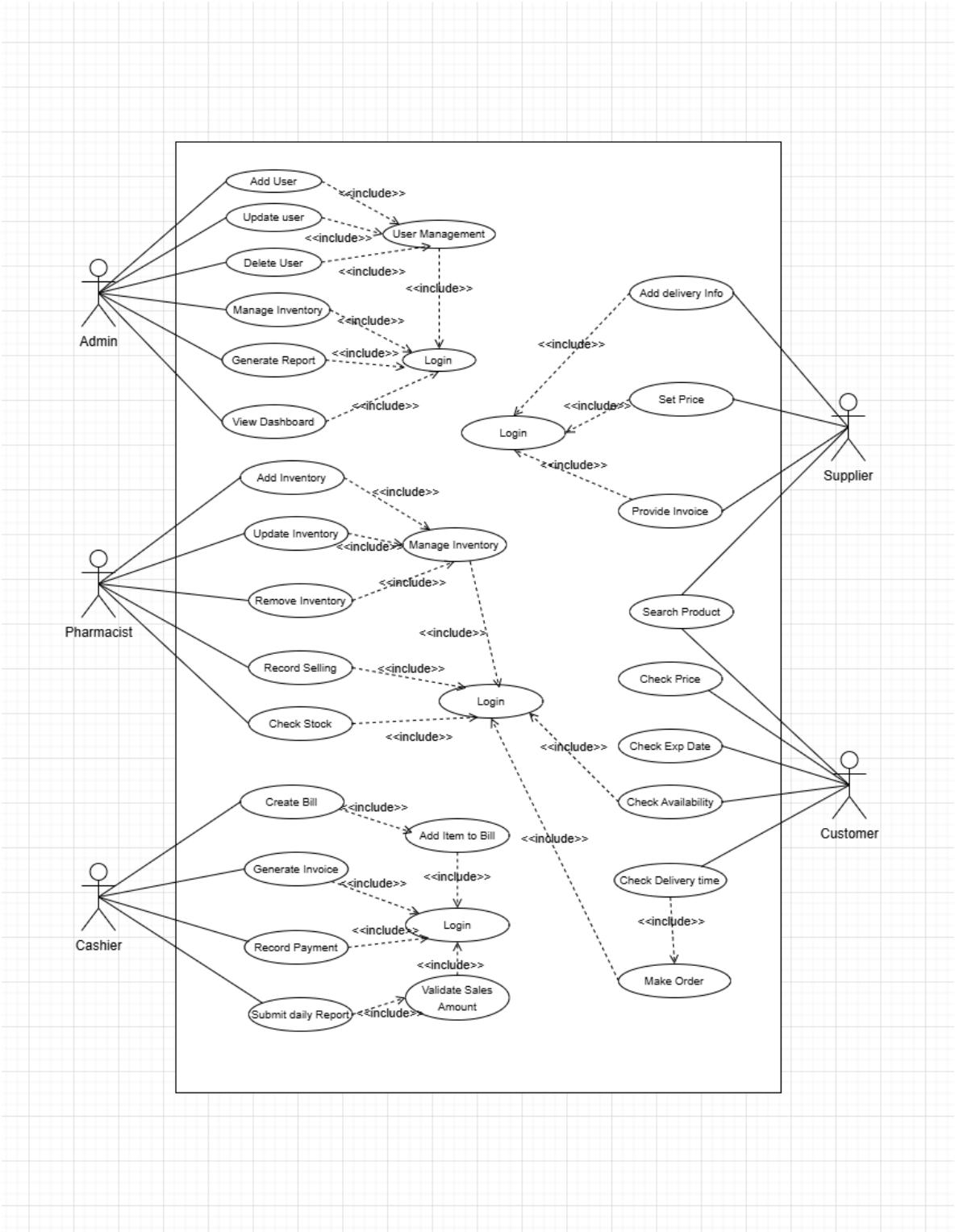
- Admin can view a dashboard with summarized reports.
- Reports include sales summaries, inventory status, expiries, supplier performance, and financial logs.
- Admin can generate printable/downloadable reports.
- Reports integrate data from all user actions for accuracy.

2.6 Scope 6: Supplier Management

Features:

- Suppliers can update the delivery status of medicines.
- Suppliers can provide invoices and add batch numbers.
- Suppliers can respond to stock requests from pharmacies.
- Suppliers can update product pricing. e) Supplier can negotiate prices when extending stock supply.

3. USE CASE DIAGRAM



4. USER STORY TABLE

As A/An	I want to	So that	Acceptance Criteria
Customer	Create an account	I can enroll myself in the system	Valid data saves and confirmation is sent
	Login with username and password	System can authenticate me	After login, dashboard loads
	View my profile details	I can see all my information	Profile table visible with edit option
	Update my information	I can correct wrong information	Updated info saved and displayed
	Change my password	My account becomes secure	Old & new password required; success message shown
	Recover my password	I do not lose access if I forget it	OTP sent and reset allowed
	View available medicines	I can check products before visiting	List of available medicines shown
	View past invoices	I can track purchases	Invoice history list displayed
As A/An	**I want to**	**So that**	**Acceptance Criteria**
Pharmacist	Add new medicine	Newly released medicine can be added	Fields required; success message
	Search for medicine	I can check quantity	List of matching medicines displayed
	Update medicine information	Everyone has updated information	Updated medicine list shown
	Remove medicine	Medicine no longer appears	Medicine removed and list updated
	View medicine list	I see updated quantities	Updated list with delete/update options
	Send stock request	I can restock medicines	Request successfully sent
As A/An	**I want to**	**So that**	**Acceptance Criteria**
Cashier	Create bills	I can process purchases	Bill generated with totals
	Generate invoice	Customer gets receipt	Printable invoice created
	Record payments	Transactions are stored	Payment logged with timestamp
	Submit sales report	Admin can review profit	Report successfully submitted
	View stock availability	I confirm items before billing	Stock list displayed

As A/An	I want to	So that	Acceptance Criteria
Supplier	Update delivery status	Pharmacy knows progress	Status updated and visible
	Provide invoices	Documentation stays complete	Invoice uploaded and saved
	Respond to stock requests	I confirm or deny orders	Response saved
	Update pricing	Pharmacy sees correct prices	New price applied
	Negotiate pricing	I offer flexible deals	Negotiation stored
As A/An	**I want to**	**So that**	**Acceptance Criteria**
Admin	Add new users	System has authorized users	User created with role
	Update user information	Records stay correct	Changes saved
	Delete users	Unauthorized access removed	User removed
	Assign roles	Control access properly	Role permissions applied
	View dashboard	Monitor all operations	Dashboard loads with analytics
	Generate reports	Audit and analysis possible	Reports downloadable (PDF/Excel)
	Approve stock updates	Inventory stays accurate	Approval logged in system

6. REQUIREMENT TRACEABILITY MATRIX

We will go into great detail about the software needs in this section. In essence, we shall classify requirements according to the users. The functionalities of the system will be implemented based on this.

6.1 Functional Requirements

6.1.1 Admin Requirements:

No	Requirement Description
1.0	User Home Page – All features accessible
1.1	Registration / Add New Users
1.1.1	Enter new user details
1.1.2	Assign roles [ref 1.1.1]
1.2	Login
1.2.1	Enter username
1.2.2	Enter password
1.2.3	Forgot password option
1.2.3.1	Recover password
1.2.3.1.1	Enter registered email/phone
1.2.3.1.2	Send OTP/link for verification [ref 1.2.3.1.1]
1.2.4	Submit login form [ref 1.2.1, 1.2.2]
1.3	Dashboard
1.3.1	View Admin Dashboard after login [ref 1.2.4]
1.4	Profile Management
1.4.1	View profile details [ref 1.3.1]
1.4.2	Update profile
1.4.2.1	Enter new contact/personal info
1.4.2.2	Submit updated info [ref 1.4.2.1]
1.5	Change Password
1.5.1	Enter old password
1.5.2	Enter new password
1.5.3	Confirm new password
1.5.4	Submit change password form [ref 1.5.1, 1.5.2, 1.5.3]
2.0	User & System Management
2.1	Update user information [ref 1.3.2]
2.2	Delete users [ref 2.1]
2.3	Assign roles & control access [ref 2.1]
2.4	Approve stock updates [ref 2.1]
2.5	Generate reports [ref 1.3.1]
2.6	View dashboard analytics [ref 1.3.1]

6.1.2 Pharmacist Requirements:

1.0	User home page – All features availability
1.1	Login [ref Customer 1.2]
1.2	Dashboard
1.2.1	Side menu
1.2.1.1	Dashboard button
1.2.1.2	Add new medicine button
1.2.1.2.1	Taking medicine name
1.2.1.2.2	Taking manufacturer
1.2.1.2.3	Taking price
1.2.1.2.4	Taking quantity
1.2.1.2.5	Submitting the form
1.2.1.2.6	Showing success message
1.2.1.3	Search medicine button
1.2.1.3.1	Taking medicine name
1.2.1.3.2	Submitting the form
1.2.1.3.3	Showing matching medicine list
1.2.1.4	Update medicine information
1.2.1.4.1	Showing existing medicine details
1.2.1.4.2	Taking updated name
1.2.1.4.3	Taking updated price
1.2.1.4.4	Taking updated quantity
1.2.1.4.5	Submitting updated information
1.2.1.4.6	Showing updated medicine list
1.2.1.5	Remove medicine button
1.2.1.5.1	Showing medicine details
1.2.1.5.2	Confirming deletion
1.2.1.5.3	Removing medicine
1.2.1.5.4	Showing updated list
1.2.1.6	View medicine list
1.2.1.6.1	Showing full list of medicines
1.2.1.6.2	Showing quantity, price, update & delete options
1.2.1.7	Send stock request
1.2.1.7.1	Taking item name
1.2.1.7.2	Taking required quantity
1.2.1.7.3	Submitting request
1.2.1.7.4	Showing confirmation message
1.2.1.8	Logout button

6.1.3 Cashier Requirements:

No	Requirement Description
1.0	User Home Page – All features accessible
1.1	Login
1.1.1	Enter username
1.1.2	Enter password
1.1.3	Forgot password option
1.1.3.1	Recover password
1.1.3.1.1	Enter registered email/phone
1.1.3.1.2	Send OTP/link for verification [ref 1.1.3.1.1]
1.1.4	Submit login form [ref 1.1.1, 1.1.2]
1.2	Dashboard
1.2.1	View Cashier Dashboard after login [ref 1.1.4]
1.3	Profile Management
1.3.1	View profile details (Name, ID, Contact) [ref 1.2.1]
1.3.2	Update profile
1.3.2.1	Enter new contact/personal info
1.3.2.2	Submit updated info [ref 1.3.2.1]
1.4	Change Password
1.4.1	Enter old password
1.4.2	Enter new password
1.4.3	Confirm new password
1.4.4	Submit change password form [ref 1.4.1, 1.4.2, 1.4.3]
2.0	Billing & Sales Operations
2.1	Create bills [ref 1.2.1]
2.2	Generate invoice [ref 2.1]
2.3	Record payments [ref 2.1]
2.4	Submit sales report [ref 2.1]
2.5	View stock availability [ref 1.2.1]

6.1.4 Supplier Requirements:

No	Requirement Description
1.0	User Home Page – All features accessible
1.1	**Login**
1.1.1	Enter username
1.1.2	Enter password
1.1.3	Forgot password option
1.1.3.1	Recover password
1.1.3.1.1	Enter registered email/phone
1.1.3.1.2	Send OTP/link for verification [ref 1.1.3.1.1]
1.1.4	Submit login form [ref 1.1.1, 1.1.2]
1.2	Dashboard

1.2.1	View Supplier Dashboard after login [ref 1.1.4]
2.0	Delivery & Stock Operations
2.1	Update delivery status [ref 1.2.1]
2.2	Provide invoices [ref 2.1]
2.3	Respond to stock requests [ref 2.1]
2.4	Update pricing [ref 2.1]
2.5	Negotiate pricing deals [ref 2.1]

6.1.5 Customer Requirements:

1.0	User home page – All features availability
1.1	**Registration**
1.1.1	Taking name
1.1.2	Taking email address
1.1.3	Taking username
1.1.4	Taking password
1.1.5	Taking confirm password
1.1.6	Taking gender
1.1.7	Taking phone number
1.1.8	Taking date of birth
1.1.9	Taking address
1.1.10	Taking postcode
1.1.11	Taking city
1.1.12	Taking division
1.1.13	Submitting the registration form
1.1.14	Resetting the form
1.1.15	Showing successful registration confirmation
1.2	**Login**
1.2.1	Taking username
1.2.2	Taking password
1.2.3	Submitting login
1.2.4	Showing dashboard after successful login
1.2.5	Forgot password button
1.2.5.1	Password recovery
1.2.5.1.1	Taking registered contact/email
1.2.5.1.2	Sending OTP
1.2.5.1.3	Taking OTP
1.2.5.1.4	Resetting password
1.2.5.1.5	Showing success message
1.3	**Dashboard**
1.3.1	Side menu
1.3.1.1	Dashboard button
1.3.1.2	View profile button
1.3.1.2.1	Show name

1.3.1.2.2	Show email
1.3.1.2.3	Show gender
1.3.1.2.4	Show phone number
1.3.1.2.5	Show date of birth
1.3.1.2.6	Show address
1.3.1.2.7	Show postcode
1.3.1.2.8	Show city
1.3.1.2.9	Show division
1.3.1.2.10	Show profile picture
1.3.1.2.11	Edit profile button
1.3.1.2.11.1	Edit profile
1.3.1.2.11.1.1	Show existing name and take updated name
1.3.1.2.11.1.2	Show existing email and take updated email
1.3.1.2.11.1.3	Show existing gender and take updated gender
1.3.1.2.11.1.4	Show existing phone number and take updated phone number
1.3.1.2.11.1.5	Show existing date of birth and take updated DOB
1.3.1.2.11.1.6	Show existing address and take updated address
1.3.1.2.11.1.7	Show existing postcode and take updated postcode
1.3.1.2.11.1.8	Show existing city and take updated city
1.3.1.2.11.1.9	Show existing division and take updated division
1.3.1.2.11.1.10	Submitting updated information
1.3.1.2.11.1.11	View profile
1.3.1.3	Change password button
1.3.1.3.1	Taking old password
1.3.1.3.2	Taking new password
1.3.1.3.3	Taking confirm new password
1.3.1.3.4	Submitting the form
1.3.1.3.5	Showing success message
1.3.1.4	View available medicines
1.3.1.4.1	Showing list of available medicines
1.3.1.4.2	Showing name, price, and availability
1.3.1.5	View past invoices
1.3.1.5.1	Showing invoice history list
1.3.1.5.2	Viewing invoice details
1.3.1.6	Logout button
1.3.1.6.1	Show login page